

## Slide 1 – Title Slide

### Speaker Notes:

- Good morning/afternoon everyone.
  - I am *[Your Name]*, and today I will be presenting our mini project titled “**TransportSense: AI-Powered Transport Emission Analytics.**”
  - This project focuses on analyzing and predicting global transport emissions using modern technologies.
  - It combines **functional programming, data analytics, and AI** to build an intelligent emission analysis platform.
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## Slide 2 – Abstract

### Speaker Notes:

- The transport sector contributes about **24% of global CO<sub>2</sub> emissions**, making it a major environmental concern.
  - However, current tools are **fragmented and lack predictive capabilities**.
  - Our system, *TransportSense*, solves this by providing:
    - Data analysis
    - Visualization
    - Forecasting
  - It uses **16 years of data across 196 countries**.
  - The backend is built using **Haskell**, and the frontend uses **vanilla JavaScript**, highlighting efficient and lightweight design.
  - Functional programming concepts like **pure functions, ADTs, and monoids** are applied to real-world analytics.
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## Slide 3 – Abstract (Continued)

### Speaker Notes:

- The backend consists of **18 Haskell modules**, each handling tasks like:
  - Emission calculation
  - Trend analysis
  - Policy simulation
- The frontend includes **8 interactive dashboards** such as:
  - Global stats

- Country comparison
  - Emission calculator
  - We also integrated an **AI chatbot using Llama 3.1**, which allows users to query data interactively.
  - Additionally, a **linear regression model** predicts emission trends up to **2035**.
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#### Slide 4 – Facts & Figures

##### Speaker Notes:

- Transport contributes about **8.7 gigatons of CO<sub>2</sub> annually**.
  - Around **75% of emissions come from road transport**, especially private vehicles.
  - If no action is taken, emissions are expected to increase by **16% by 2050**.
  - Current tools only show past data and do not support:
    - Forecasting
    - Policy simulation
  - This highlights the **need for smarter analytical tools like TransportSense**.
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#### Slide 5 – Facts & Figures (Continued)

##### Speaker Notes:

- Switching from cars to **rail or cycling can reduce emissions by 60–90%**.
  - EV adoption varies significantly between countries, so **localized analysis is essential**.
  - Although data is available, it is mostly in **CSV or PDF formats**, making it difficult to use.
  - Therefore, we need an **interactive and user-friendly platform**, which our system provides.
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#### Slide 6 – Literature Review

##### Speaker Notes:

- We studied various research works:
  - Creutzig et al. focused on **transport decarbonization strategies**.
  - IEA reports highlighted transport as the **hardest sector to reduce emissions**.
  - Rodrigue explained **transport systems and sustainability metrics**.
  - Brand et al. worked on **future transport scenarios using modeling**.
- These studies helped us understand the domain and design our system.

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## Slide 7 – Literature Review (Continued)

### Speaker Notes:

- We also referred to technical works:
    - Hudak and O’Sullivan explained **functional programming with Haskell**.
    - Pollet et al. discussed **dashboard design for climate data**.
    - Meta AI introduced **Llama 3**, which we used for our chatbot.
  - These references guided both our **technical implementation and design**.
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## Slide 8 – Research Gaps

### Speaker Notes:

- Existing systems have several limitations:
    1. No **predictive analytics**
    2. No **custom dataset import**
    3. AI tools often **hallucinate incorrect data**
    4. No use of **functional programming for such systems**
  - Our project aims to **bridge all these gaps** effectively.
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## Slide 9 – Novelty of Proposed System

### Speaker Notes:

- Our system introduces several innovations:
    - First use of **Haskell for emission analytics**
    - AI chatbot that gives **dataset-specific answers**, avoiding hallucination
    - **Client-side dataset import**, no server needed
    - A custom-built **linear regression engine**
  - These features make our system **unique and efficient**.
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## Slide 10 – System Design and Modules

### Speaker Notes:

- The system has four main components:
  1. **Backend (Haskell)**

- 18 modules handling data processing
  - 2. **Frontend (JavaScript)**
    - Interactive dashboards with charts
  - 3. **AI Prediction Engine**
    - Linear regression + chatbot
  - 4. **Deployment**
    - Docker, Railway, Firebase
  - This modular design ensures **scalability and maintainability**.
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## Slide 11 – Contributions

### Speaker Notes:

- Our contributions include:
    - A fully functional **Haskell-based backend**
    - A **global dashboard** covering 196 countries
    - **Policy simulation models** for emission reduction
    - A **context-aware AI chatbot**
    - A **custom forecasting system** and dataset importer
  - Overall, the project integrates **analytics, AI, and functional programming**.
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## Slide 12 – References

### Speaker Notes:

- These are the sources we referred to, including:
    - IEA reports
    - Research papers
    - Haskell documentation
    - Chart.js and Llama 3 resources
  - These helped ensure our project is **technically strong and research-backed**.
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## Slide 13 – Thank You

### Speaker Notes:

- Thank you for your attention.

- We hope this presentation gave you a clear understanding of our system.
- I would be happy to answer any questions.